Portable Embedded Deployment of 5G Standalone (SA) System with Software-Defined Radio

Followup on Paper "Minimal Open-Source 5G-SA Campus Reference System"

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***Abstract*—We present a low-cost 5G-SA system based on edge-computing hardware with open-source components and off-the-shelf hardware in this paper. Our system is suitable for nomadic and ad-hoc 5G-SA use cases that require low-power consumption and small space. We show a complete development covering how to prepare the operating system for the installation of the 5G next-generation node B (gNB) and how to install and deploy the 5G-SA software components. We evaluate the performance, indoor coverage range, and power consumption of our system with measurements. We show that our system achieves acceptable data rates, and is very space and energy-efficient with an average power consumption of ≈ 24.81 W. We provide detailed and maintained documentation on the deployment of our system and discuss possible extensions for future work.**

***Index Terms*—5G, NR, Standalone (SA), Edge Computing, Campus Network, Software Defined Radio (SDR)**

I. Introduction

The fifth generation of mobile communications stands for consistent modularization and virtualization of systems and opens the technology for new fields of application in industry, research, and the Internet of Things [1]–[4]. Open-source 5G systems and off-the-shelf hardware play a special role in industrial and private networks — especially in terms of scalability and costs [5]–[7]. The performance and stability of the open-source and SDR-based systems are constantly improving [8]–[10].

The fifth generation expands mobile networks with nomadic, even mobile, and Ad-hoc structures and the concept of side link communication to cover new use cases, such as V2V communication, IoT applications, or ad hoc infrastructure in emergencies [11]–[13]. Low-power consumption and compact dimensions and low weight are requirements of the overall system in these new nomadic and Ad-hoc 5G-SA use cases.

With the published paper on a 5G-SA open-source reference system, a cost-effective deployment was presented, and attached instructions were constantly adapted and extended to the current state of development [8]. This paper is a supplementary report on the original work and presents our current 5G-SA open-source-based deployment for a 5G-SA system on edge-computing hardware. This current deployment achieves acceptable data rates, while being very space-saving and efficient, with an average power consumption of ≈ 24.81 W. The deployment is designed for 5G research, and development activities. Furthermore, the system is applicable for nomadic and Ad-hoc 5G-SA system use cases.

The contributions of this paper:

- We provide brief instructions on preparing the operating system of edge-computing hardware for the 5G-System.
- We present the development of a 5G-System based on a Seeed Studio NVIDIA© Jetson AGX Orin™ 32 GB H01 Kit.
- We show performance, indoor coverage range, and power consumption measurements of our deployment.

The paper is structured as follows. First, we provide an overview of applied hardware and software components, to enable others to deploy our proposed 5G-SA system as a reference. Then, we describe the necessary operating system kernel adaptions on our edge-computing hardware for the 5G next-generation node B (gNB) installation. The fourth section presents some details of the deployment of the 5G-SA software components. Finally, we present the results of the performance and power consumption measurements and conclude our work.

II. Hardware- and Software Setup and Configuration Parameters

We present our hardware and software components in detail in this section. We provide this information to enable new 5G developers to take our system as a reference for their deployment and enhance it.

Table I lists the components of our deployment. The deployment is feasible on different edge-computing systems, with the adoption of the gNB configuration (refer Section IV-B).

Table II lists the software components of our 5G system. The components in the provided version are interoperable, and we used this deployment for the tests in Section V. We continuously maintain the documentation in our deployment repository[1] to test the latest supported features of the software components.

[1]https://www.seeedstudio.com/AGX-Orin-32GB-H01-Kit-p-5569.html
[2]https://www.ettus.com/all-products/usrp-b205mini-i/

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TABLE I
HARDWARE COMPONENTS OF OUR 5G GNB AND CORE SYSTEM.

Parameter	Value
Computer - Seeed Studio NVIDIA© Jetson AGX Orin™ [1]	
CPU	8-core NVIDIA Arm© Cortex A78AE v8.2 64-bit CPU @ 2.2 GHz
RAM	32 GB LPDDR5
Storge	64 GB eMMC
Ethernet	10 GbE
USB	2x USB 3.2 Type-A
SDR - Ettus USRP™ B205mini-i [2]	
Frequency Range	70 MHz – 6 GHz
Bandwidth	56 MHz
USB	1x USB 3.0
FPGA	Xilinx Spartan-6 XC6SLX150
Antenna - Delock 5G SMA plug Antenna	
Frequency Range	3.3 GHz to 5 GHz
max. Gain	≈ 3 dBi @ 3.75 GHz

TABLE II
SOFTWARE OF OUR 5G SYSTEM.

Parameter	Value
Operating System	
Linux	Ubuntu 20.04 (Focal)
Jetpack	5.1.1
L4T Version	35.3.1
Kernel	5.10.104-tegra (customized, SCTP added)
open5GS core	
Tag	v2.7.0
Git	https://github.com/open5gs/open5gs.git
Database	MongoDB v6.0
USRP™ Hardware Driver (UHD™)	
Tag	v4.6.0
Git	https://github.com/EttusResearch/uhd.git
srsRAN project gNB	
Tag	release_23_10_1
Git	https://github.com/srsran/srsRAN_Project.git
CoSA 5G Deployment Tutorials and Scripts	
Tag	v1.1
Git	https://git.mylab.th-luebeck.de/5g/cosa-5g-private-campus-network-deployment-oai.git

We designed mounting adapters, to handle the Seeed Studio NVIDIA© Jetson AGX Orin™ with the Ettus USRP™ B205mini-i as a compact 5G system. Figure 1 shows the schematic of the system. The mounting adapters are screwed, using the mounting holes of the USRP and the Jetson. A gap was created between the SDR and the Jetson to improve air circulation and cooling.

Figure 2 illustrates our 5G-SA system on the embedded computer Jetson AGX Orin™. The B205mini-i SDR with

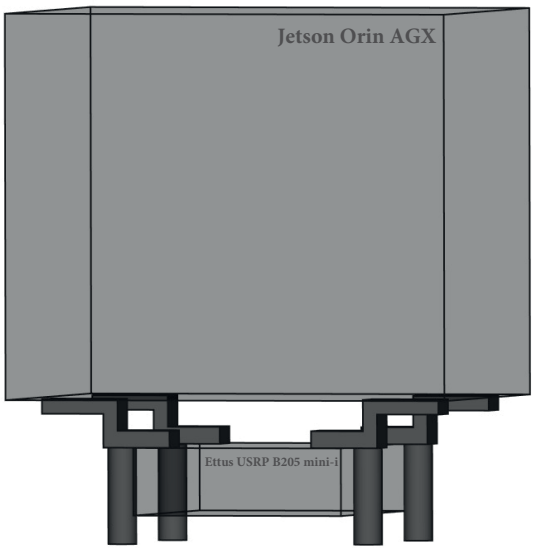

Fig. 1. Designed Mounting System for Ettus USRP™ B205mini-i on the Seeed Studio NVIDIA© Jetson AGX Orin™.

the plug antennas from Delock is the RF front end of our gNB. We attach an Eaton mini uninterruptible power supply to operate our system independently from the main power line. The system is highly portable and has a very small form factor.

III. PREPARATION OF THE OPERATING SYSTEM OF THE EDGE-COMPUTING HARDWARE

Due to the limited and specialized hardware components of the NVIDIA© Jetson™ family their operating systems (OS), respectively their kernels, are customized. However, deploying a 5G-SA system requires modules not included in the delivered kernel. Therefore, we provide brief instructions on customizing the kernel exemplary for a Seeed Studio NVIDIA© Jetson AGX Orin™.

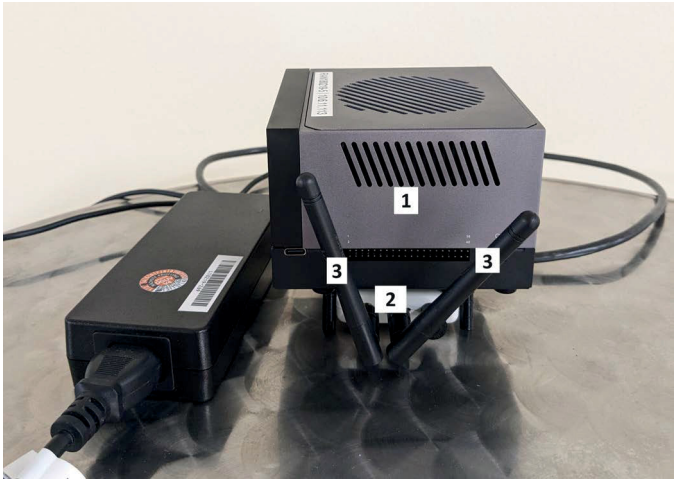

Fig. 2. **5G-SA System on an Embedded Computer.** 1: Jetson AGX Orin™ | 2: Ettus USRP™ B205mini-i | 3: Delock 5G SMA plug antennas.

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Our Jetson AGX Orin™ runs with Jetpack 5.1.1 and L4T 35.3.1 with the kernel 5.10.104-tegra. The kernel lacks a module required to support the Stream Transmission Control Protocol (SCTP). In the following, we show how we customized and built the kernel on the target device. For a detailed description, we refer to our deployment repository [3].

First, we retrieve the kernel sources in version 35.3.1 from the *Jetson Linux Archive*. The kernel configuration is customized with the menu application started by `make menuconfig`. We select the *SCTP-Protocol* as module [M] in *Networking support* → *Networking options*. Then, in the configuration file `.config` the parameter `CONFIG_LOCALVERSION="-tegra-sctp"` is adapted. The kernel build of this customized kernel is performed with `sudo make -j5 Image` for the Image file and `sudo make modules` and `sudo make modules_install` for the modules. Finally, the compiled `Image` file is copied to `/boot/Image`, whereby the original `/boot/Image` was backed up before.

The rebooted system starts with the customized kernel `5.10.104-tegra-sctp`. `5.10.104-tegra-sctp` is tested with `sctp_darn`, which is supposed to start the SCTP-listener. The original kernel `5.10.104-tegra` can be reactivated, by replacing the `/boot/Image` file with the previously created backup `Image`.

We activate `Jetson Clocks` and `Max Power Mode` (max. 60 W) using `jtop`, to achieve the required performance on the Edge-Computer. With the customized kernel and performance measures, the Jetson is prepared for deploying and operating a 5G system in a box. The deployment is shown in the next section.

IV. 5G System Deployment

A 5G system comprises the Radio Access Network (RAN), 5G-Core, and the Service, Management, and Orchestration (SMO) functions. We propose a (minimal) deployment in a box with core and gNB. The SMO is not required for system operation and therefore not considered in this deployment.

We documented our deployments with the Core and RAN from Open Air Interface, open5GS Core, and srsRAN in our GITLab repository[4]. This work presents a 5G deployment with the open5Gs core and srsRAN. We use shell scripts from our deployment repository[2], to configure and run the 5G components. The services are configured to start on boot automatically. In the following subsections, we describe the specifics for 5G-SA deployment on a Jetson AGX Orin™. Our detailed and constantly updated tutorial is available as a GITLab repository[1].

A. 5G Core Deployment

This section describes the deployment of the open5GS core on the Jetson AGX Orin™. Our Jetson with customized kernel runs under Ubuntu focal (v20.04). To ensure a stable 5G-SA system deployment, we build the packages from sources in fixed versions. Our deployment repository[1] documents a set of compatible versions and configurations to get started with a 5G-SA system and is adaptable to customizations.

Our open5GS core in version 2.7.0 is installed with mongoDB in version 6.0. We adopt the PLMN (MCC and MNC), TAC, and Slices (SST and SD) in the configuration file `sample.yaml` to our approaches. Finally, we insert the SIM-card credentials to the database of the 5G-core with the command line tool `open5gs-dbctl`. Then, we start the prepared 5G-SA core. We observe connected gNBs and UEs in the console log output.

B. 5G RAN Deployment - gNB

We applied the srsRAN project at release_23_10_1 with UHD™ v4.6.0.0 to create the physical layer with an SDR. For the execution of the gNB software on the Jetson, we use a modified configuration file. Compared to prior deployments, we now use the `QUAM256` modulation and coding scheme (MCS). Furthermore, the target block error rate (`olla_target_bler`) in the physical uplink and downlink shared channels (PUSCH and PDSCH) were adapted to 5 %.

To overcome unstable latency behavior, we adapted the TDD slot configuration to `DDSU` with 8 `D` symbols and 0 `U` symbols in the `S` slot. The slot configuration consists of 4 Slots with a resulting periodicity of 2 ms. Finally, we adapted the scheduling request period (`sr_period_ms`) in the physical uplink control channel (PUCCH) to 2 ms to minimize the latency. We increased the overall performance of our deployment with these adoptions.

The CPU resources of the Jetson AGX Orin™ limit the maximum possible bandwidth of the 5G system. At 40 MHz bandwidth, we observe too few resources for the signal processing by continuous `overflow` and `underflow` messages in the console output of the srsRAN gNB. Therefore, we decreased the bandwidth to 30 MHz applying a sampling rate of 46.08 MHz. We observed `overflow` and `underflow` infrequently for this configuration, but the system behaves stably.

V. Performance Measurements

We measured the system's data throughput, latency, indoor coverage range, and power consumption to demonstrate the capabilities of our 5G-SA system. We selected a Raspberry Pi 4B with a Quectel RM520N-GL modem as User equipment (UE) to perform the measurements. We recorded 119 throughput and ping round trip times (RTT) measurements over ≈ 100 min. Our 5G system and the UE were placed in an empty (no person) office room with a distance of 1 m. The throughput was measured with *iperf3*. The Raspberry Pi was the only connected UE in the network during the performance measurements. We used a local iperf server on the Jetson Orin AGX and measured the TCP throughput for the uplink and downlink (reverse mode). The average throughput

[3] https://git.mylab.th-luebeck.de/5g/cosa-5g-private-campus-network-deployment-oai/-/tree/main/5G_on_seeed_jetson_orin_agx

[4] https://git.mylab.th-luebeck.de/5g/cosa-5g-private-campus-network-deployment-oai

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measurements were taken over 15 s, where the first 5 s were omitted due to the TCP slow start. After each uplink and downlink measurement, the ping RTT measurement was performed. The results were transmitted via MQTT and NodeRed to an InfluxDB for further post-processing and visualization.

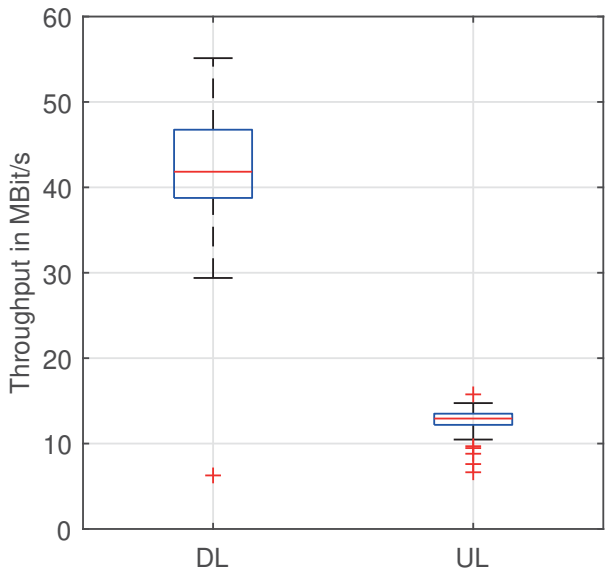

Fig. 3. Results for iperf measurements between UE and 5G-Gateway.

Figure 3 illustrates the results of the `iperf` measurement via TCP. During the measurement, we observed varying channel quality index (CQI) values, between 11 and 15. Therefore, the MCS values varied between 15 and 19 for the downlink and between 10 and 15 for the uplink. We calculate an expected downlink of 43.9 Mbit/s and uplink data rate of 14.5 MBit/s (referring to the throughput calculation[5]), with these conditions.

The maximum possible MCS of 28 was not reached. With a median downlink data rate of 41.83 Mbits/s, our deployment shows an expected performance for a single input single output (SISO) system with 30 MHz bandwidth and the given channel quality. The median uplink data rate was 12.93 Mbits/s and is also in line with the throughput calculation.

Figure 3 also indicates an outlier in the downlink and few outliers in the uplink measurements. The system remained stable during the measurements.

The latency RTT of our 5G-SA system was measured with `ping` between the UE and the 5G-SA Package Gateway (PGW)/User plane functions (UPF). The median RTT of our system is 16.75 ms with a minimum of 9.71 ms. Ultra-low latency with latencies $<$ 1 ms as defined by the 3GPP are not reached with the proposed development [14], [15].

[5]https://5g-tools.com/5g-nr-throughput-calculator/

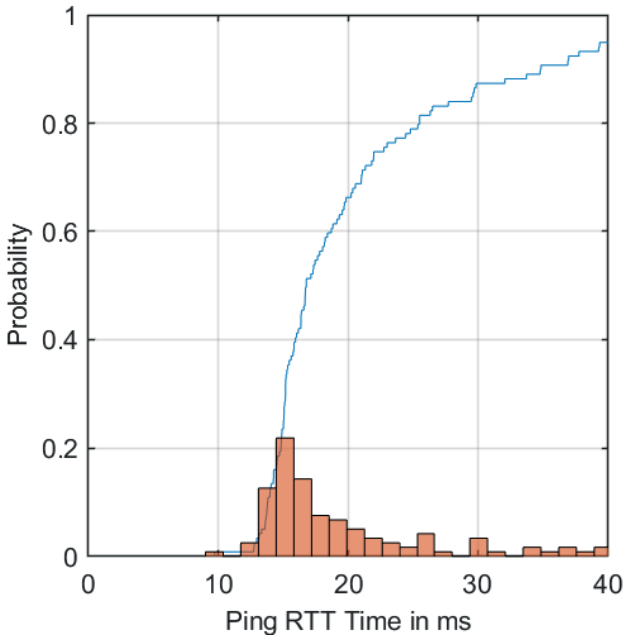

Fig. 4. Results for ping (RTT) measurements between UE and 5G-Gateway.

Figure 4 illustrates the results of the RTT measurement as a histogram with an ECDF. 66 % of the measured RTT's were lower than 20 ms and 87 % were lower than 30 ms.

We also performed measurements in a corridor with a length of $\approx$ 50 m. A power amplifier (ZVA-183-S+) and a low noise amplifier (ZX60-123LN-S+) were integrated into the SDR and directional antennas (POYNTING A-HELI-0031-V2-01) were used. The transmission power of the system was 26.87 dBm ($\approx$ 500 mW). The measurement setup is shown in Figure 5. For distances between 10 m and 50 m, the maximum uplink and downlink data rate and the RSRP value were recorded at a distance of 5 m using the same measurement setup as in the office measurement. The results are shown in Figure 6.

The uplink (up to 15 Mbit/s) and downlink (up to 50 Mbit/s) data rates are comparable to the values measured in the office. At a distance of 25 m to 35 m, both RSRP and data rates are reduced. This can be attributed to unfavorable interference in the corridor. In addition, a fire door is installed in the corridor at a distance of $\approx$ 30 m, which was open during the measurement, but influences the signal propagation (additional

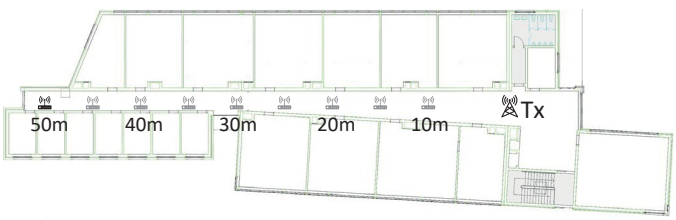

Fig. 5. Measurement setup for UE at several distances to the 5G system with amplifiers.

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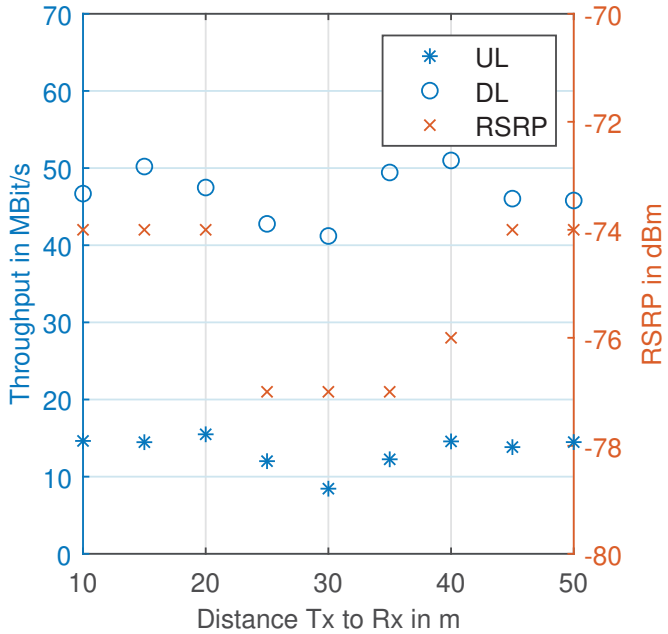

Fig. 6. Measurement results for UE at several distances to the 5G system with amplifiers.

reflections). The data rates and signal strengths were within the expected range, even at the maximum distance of 50 m.

Finally, we performed measurements on the power consumption of our 5G-SA system. Therefore, we connected one UE to our system and generated continuous data traffic with an `iperf` measurement for a duration of 3600 s. The average power consumption of our system is ≈ 24.81 W. Application scenarios, such as nomadic or Ad-hoc 5G-SA networks with battery-based power supply, require low-power consumption to increase their manageability. These measurement results provide a basis for the dimensioning of the energy source for a particular application scenario.

VI. Conclusion and Future Work

This work presents the development of a portable, low-power complete 5G-SA system. Our deployment combines 5G core and gNB in a box but is not limited to this configuration. We demonstrate the preparation and installation of open-source 5G components on an embedded computer with connected SDR as a radio frontend. Our detailed and maintained documentation on the deployment is provided with the supplementary repository[1]. Our performance measurements are comparable with the calculated throughput rates for the given bandwidth and channel quality. Further tuning of the transmitter and receiver gain can increase the CQI and throughput of the system. Furthermore, the range of the SDR-based system is extendable with external power amplifiers for the transmitter and low-noise amplifiers for the receiver. Our measurements demonstrate coverage up to 50 m with expected data rates for an indoor scenario. The evaluation of a complex indoor scenario with signal propagation through walls and reflections is outstanding. Due to the low-power consumption of ≈ 24.81 W, the proposed system is operable with batteries and applies to nomadic and Ad-hoc 5G-SA use cases.

Our measurements demonstrate the performance of the 5G-SA system. We will extend the system in a multicell setup with several gNBs running on edge-computing embedded computers connected to the 5G core. With the availability of side link functionality in the open-source project, this setup is suitable to act as a range extender in nomadic 5G networks.

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References

[1] Y. Zhang, *Network Function Virtualization: Concepts and Applicability in 5G Networks*. John Wiley & Sons, 2018.

[2] M. Agiwal, H. Kwon, S. Park, and H. Jin, "A survey on 4g-5g dual connectivity: Road to 5g implementation," *IEEE Access*, vol. 9, pp. 16 193–16 210, 2021.

[3] C. Jenal, S. Endreß, O. Kühne, and C. Zylka, "Technological transformation processes and resistance; on the conflict potential of 5g using the example of 5g network expansion in germany," *Sustainability*, vol. 13, no. 24, 2021. [Online]. Available: https://www.mdpi.com/2071-1050/13/24/13550

[4] P. Varga, J. Peto, A. Franko, D. Balla, D. Haja, F. Janky, G. Soos, D. Ficzere, M. Maliosz, and L. Toka, "5g support for industrial iot applications— challenges, solutions, and research gaps," *Sensors*, vol. 20, no. 3, 2020. [Online]. Available: https://www.mdpi.com/1424-8220/20/3/828

[5] J. Ordonez-Lucena, J. F. Chavarria, L. M. Contreras, and A. Pastor, "The use of 5g non-public networks to support industry 4.0 scenarios," in *2019 IEEE Conference on Standards for Communications and Networking (CSCN)*. IEEE, 2019, pp. 1–7.

[6] X. Xiong, W. Xiang, K. Zheng, H. Shen, and X. Wei, "An open source sdr-based noma system for 5g networks," *IEEE Wireless Communications*, vol. 22, no. 6, pp. 24–32, 2015.

[7] A. Ibanez, J. Sanchez, D. Gomez–Barquero, J. Mika, S. Babel, and K. Kuehnhammer, "5g broadcast sdr open source platforms," in *2022 IEEE International Symposium on Broadband Multimedia Systems and Broadcasting (BMSB)*, 2022, pp. 01–06.

[8] F. John, J. Schuljak, B. Sievers, L. Vosteen, A. Hanemann, and H. Hellbrück, "A Reference Deployment of a Minimal Open-Source Private Industry and Campus 5G Standalone (SA) System," in *2022 IEEE 10th International Conference on Information, Communication and Networks (ICICN)*, Aug 2022, pp. 1–9.

[9] O. Kaestner, T. Bojer, C. Christoph, J. Dreyer, and R. Toenjes, "Performance evaluation of sdr-based 5g networks," in *Mobile Communication - Technologies and Applications; 27th ITG-Symposium*, 2023, pp. 104–109.

[10] F. John, B. Sievers, O. H. Sellhorn, and H. Hellbrueck, "Two industrial reference demonstrators for high throughput and low latency in 5g-standalone (sa) network setups," in *Mobile Communication - Technologies and Applications; 27th ITG-Symposium*, 2023, pp. 110–115.

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[11] Ö. Bulakci, Z. Ren, C. Zhou, J. Eichinger, P. Fertl, D. Gozalvez-Serrano, and S. Stanczak, "Towards flexible network deployment in 5g: Nomadic node enhancement to heterogeneous networks," in *2015 IEEE International Conference on Communication Workshop (ICCW)*, 2015, pp. 2572–2577.

[12] H. Tullberg, Z. Li, A. Hoglund, P. Fertl, D. Gozalvez-Serrano, K. Pawlak, P. Popovski, G. Mange, and Ö. Bulakci, "Towards the metis 5g concept: First view on horizontal topics concepts," in *2014 European conference on networks and communications (EuCNC)*. IEEE, 2014, pp. 1–5.

[13] J. Gruner, A. Bloße, M. Rockstroh, T. Neumuth, and P. Rostalski, "Requirement analysis for an aerial relay in emergency response missions," *Current Directions in Biomedical Engineering*, vol. 6, no. 3, pp. 16–19, 2020.

[14] H. Ji, S. Park, J. Yeo, Y. Kim, J. Lee, and B. Shim, "Ultra-reliable and low-latency communications in 5g downlink: Physical layer aspects," *IEEE Wireless Communications*, vol. 25, no. 3, pp. 124–130, 2018.

[15] 3rd Generation Partnership Project, "Release 16 Description; Summary of Rel-16 Work Items," 3GPP, Tech. Rep., 2019.

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